

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Technology (CSE)**

QUIZ #3 (SET-A)

SUMMER SEMESTER, 2023-2024

DURATION: 20 MINUTES

FULL MARKS: 15

## CSE 4711: Artificial Intelligence

**Programmable calculators are not allowed.**

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. Consider a grid world Markov Decision Process (MDP) shown in Figure 1 with deterministic actions (all actions chosen will be executed, unless the agent is blocked by a wall). The starting state is “A” and the terminal state is “I” with a reward of 10. Assume that the reward is achieved as soon as the terminal state is reached ( $V_0^*(I) = 10$ ). The action space  $\mathcal{A}:\{\text{left, right, up, down, stay}\}$ . The discounting factor  $\gamma = 0.5$ , and there is no living reward. Applying value iteration, one can find the values of the states.

|   |   |   |
|---|---|---|
| A | B | C |
| D |   | E |
| F | G | H |
|   | I |   |

**Figure 1:** A simple grid world for Question 1.

- a) What are the values for:  $V_1^*(A)$ ,  $V_1^*(G)$ ,  $V_1^*(I)$ ,  $V^*(A)$ ,  $V^*(G)$ , and  $V^*(F)$ ?

3 + 3  
(CO1)  
(PO1)

**Solution:**

- $V_1^*(A) = 0$
- $V_1^*(G) = 0 + \gamma \times 10 = 5$
- $V_1^*(I) = 10$
- $V^*(A) = \gamma^4 \times V^*(I) = 0.625 \rightarrow \text{takes 4 steps}$
- $V^*(G) = \gamma^1 \times V^*(I) = 5 \rightarrow \text{takes 1 step}$
- $V^*(F) = \gamma^2 \times V^*(I) = 2.5 \rightarrow \text{takes 2 steps}$

**Rubric:**

- 1 point for correctly calculating each value.

- b) Assume that you have established a new function  $I(s, s')$  which returns the actions needed to go from  $s$  to  $s'$  in a deterministic setting (not a chain of actions, rather all possible singular cases). Find the values for  $I(A, B)$ ,  $I(D, E)$ , and  $I(C, C)$ .

3  
(CO1)  
(PO1)

**Solution:**

- $I(A, B) = \{right\}$
- $I(D, E) = \emptyset \rightarrow$  No singular action possible
- $I(C, C) = \{stay, up, right\}$

**Rubric:**

- 1 point for each correct answer.

2. Assume that you have \$100 and you are passing through Mohammadpur. There is an 20% chance that your money gets stolen. Your utility function is  $U(x) = \log_e(1 + x)$ , where  $x$  is the amount of money you have left.

a) What is your expected utility after going through Mohammadpur?

2  
(CO1)  
(PO1)

**Solution:**

$$0.8 \times \log_e(1 + 100) + 0.2 \times \log_e(1 + 0) = 3.6921$$

**Rubric:**

- 2 points for correctly calculating the value.

b) If you pay an premium of \$30 to an insurance company, they will reimburse you \$70 if your money gets stolen. Should you take this offer from the insurance company?

4  
(CO1)  
(PO1)

**Solution:**

Whether your money gets stolen or not, you will always have to pay \$30 to the insurance company. Your new expected utility would be:  $0.8 \times \log_e(1 + 70) + 0.2 \times \log_e(1 + 70) = 4.2627$ , which is greater than the expected utility of not taking the insurance. Thus, you should take the insurance offer.

**Rubric:**

- 3 points for correctly calculating the value.
- 1 point for correctly choosing.